

COSTA MESA WAYNE, CA, USA

WMO: 722977

Lat: 33.680N

Lon: 117.866W

Elev: 54

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 93184

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	42.8	44.9	6.9	7.8	67.1	14.7	11.6	67.4	21.3	62.6	18.5	63.8	1.3	30	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.2	89.8	65.3	86.3	65.3	83.2	64.9	71.3	82.3	70.1	80.3	69.1	78.6	7.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.7	102.1	76.4	66.3	97.0	75.3	65.3	93.7	74.3	35.1	82.2	34.0	80.3	33.2	78.8	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	13.1	12.0	DB	38.0	99.0	2.5	5.0	36.3	102.7	34.8	105.6	33.4	108.4	31.6	112.1	
			WB	33.1	73.5	2.2	1.5	31.5	74.6	30.2	75.5	29.0	76.3	27.4	77.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	58.0	57.9	60.0	62.0	64.9	68.0	72.2	73.1	72.4	68.5	62.9	57.4
	DBStd	7.12	5.12	4.83	4.31	4.35	3.78	3.00	3.38	3.37	4.61	4.80	4.93	4.34
	HDD50	5	2	1	0	0	0	0	0	0	0	0	0	2
	HDD65	1106	227	205	166	110	42	4	0	0	0	16	97	239
	CDD50	5407	249	222	310	361	462	539	688	715	672	572	386	231
	CDD65	1033	9	6	11	21	39	93	223	250	222	123	33	3
	CDH74	5385	134	93	143	237	249	228	841	1114	1211	781	308	46
	CDH80	1525	26	22	37	79	100	45	164	256	409	287	94	6
(14) Wind	WSAvg	4.8	3.8	4.4	4.9	5.6	6.1	6.1	5.9	5.3	4.7	4.1	3.5	3.6
(15) Precipitation	PrecAvg	10.6	2.4	2.7	2.2	0.8	0.1	0.0	0.0	0.1	0.4	0.2	1.2	1.5
	PrecMax	26.1	7.9	12.4	10.0	4.1	0.5	0.5	0.2	1.7	1.9	1.0	5.3	4.2
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.9	2.3	3.0	2.3	1.1	0.1	0.1	0.0	0.4	0.6	0.2	1.2	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.7	83.6	84.6	89.0	90.5	85.8	88.1	89.9	95.1	94.2	88.4	79.2
		MCWB	55.9	56.8	58.1	60.2	59.8	67.9	69.4	68.2	68.9	62.9	60.0	53.8
	2%	DB	77.6	76.2	77.7	80.8	80.6	79.2	83.9	86.1	89.0	87.5	82.1	73.4
		MCWB	53.7	54.9	56.0	57.8	61.2	66.3	68.4	68.7	69.0	61.6	57.5	54.8
	5%	DB	73.2	71.4	72.3	74.4	74.2	75.9	81.2	82.5	84.4	82.1	76.8	69.9
		MCWB	53.6	54.7	55.8	57.4	62.3	65.1	68.5	68.8	68.1	62.2	56.9	53.7
	10%	DB	68.5	66.7	68.0	69.9	70.9	73.5	78.7	79.9	80.7	77.5	72.2	66.1
		MCWB	53.2	54.4	56.3	57.8	61.0	64.0	67.8	68.3	67.6	62.5	57.1	53.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.3	61.8	62.8	63.5	66.3	69.1	72.6	72.5	73.1	70.6	64.2	62.3
		MCDB	69.9	70.2	74.9	76.9	77.2	80.3	83.4	83.9	86.7	80.7	74.9	69.3
	2%	WB	59.4	59.6	60.6	61.4	64.4	67.3	70.6	70.9	71.1	68.1	62.5	59.8
		MCDB	65.9	67.0	69.3	72.0	74.1	78.0	79.6	81.5	83.8	78.3	71.6	65.6
	5%	WB	58.1	58.2	59.2	59.9	62.8	66.0	69.4	69.7	69.9	66.3	61.3	58.4
		MCDB	64.8	65.2	67.1	69.5	71.4	75.0	78.4	79.6	80.4	75.9	69.7	64.0
	10%	WB	56.7	56.9	57.9	58.8	61.7	64.8	68.4	68.7	68.7	64.8	60.0	57.1
		MCDB	63.6	63.6	64.9	67.3	69.4	72.3	76.9	78.0	78.3	73.5	68.0	62.8
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	16.3	14.9	14.3	12.2	11.5	12.3	13.2	14.5	15.7	17.9	17.6
		MCDBR	27.0	25.3	24.5	24.3	19.6	15.4	15.8	17.3	20.8	26.0	26.8	24.5
		MCWBR	11.8	11.3	10.6	9.7	7.9	6.2	5.7	6.2	7.5	10.2	11.5	12.5
	5% WB	MCDBR	18.4	17.5	16.3	16.9	15.4	15.0	14.2	15.3	17.8	18.5	18.5	16.0
		MCWBR	10.0	9.3	8.4	7.9	6.7	6.3	5.4	6.0	7.0	8.5	9.5	9.7
(40) Clear-Sky Solar Irradiance	taub		0.323	0.333	0.357	0.363	0.362	0.377	0.383	0.368	0.353	0.354	0.312	0.311
	taud		2.537	2.522	2.430	2.420	2.454	2.416	2.466	2.493	2.526	2.499	2.609	2.611
	Ebn at Noon		283	292	292	293	293	287	284	288	289	280	286	283
	Edh at Noon		26	30	35	37	36	37	35	34	31	30	24	23
(44) All-Sky Solar Radiation	RadAvg		1016	1295	1729	2100	2246	2357	2355	2244	1896	1440	1123	919
(45)	RadStd		69	126	91	107	154	147	124	64	77	105	68	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.42	N/A	N/A	+3.38	N/A	N/A	N/A	-284	+558	+306
(47) Regional (50 neighbors)	+1.13	N/A	-3.06	+1.30	N/A	N/A	N/A	-202	+401	+202

Nomenclature: See separate page